

OPAMP

- Operational Amplifiers, or Op-amps as they are more commonly called, are one of the basic building blocks of Analogue Electronic Circuits
- An **Operational Amplifier**, or op-amp for short, is fundamentally a voltage amplifying device designed to be used with external feedback components such as resistors and capacitors between its output and input terminals.

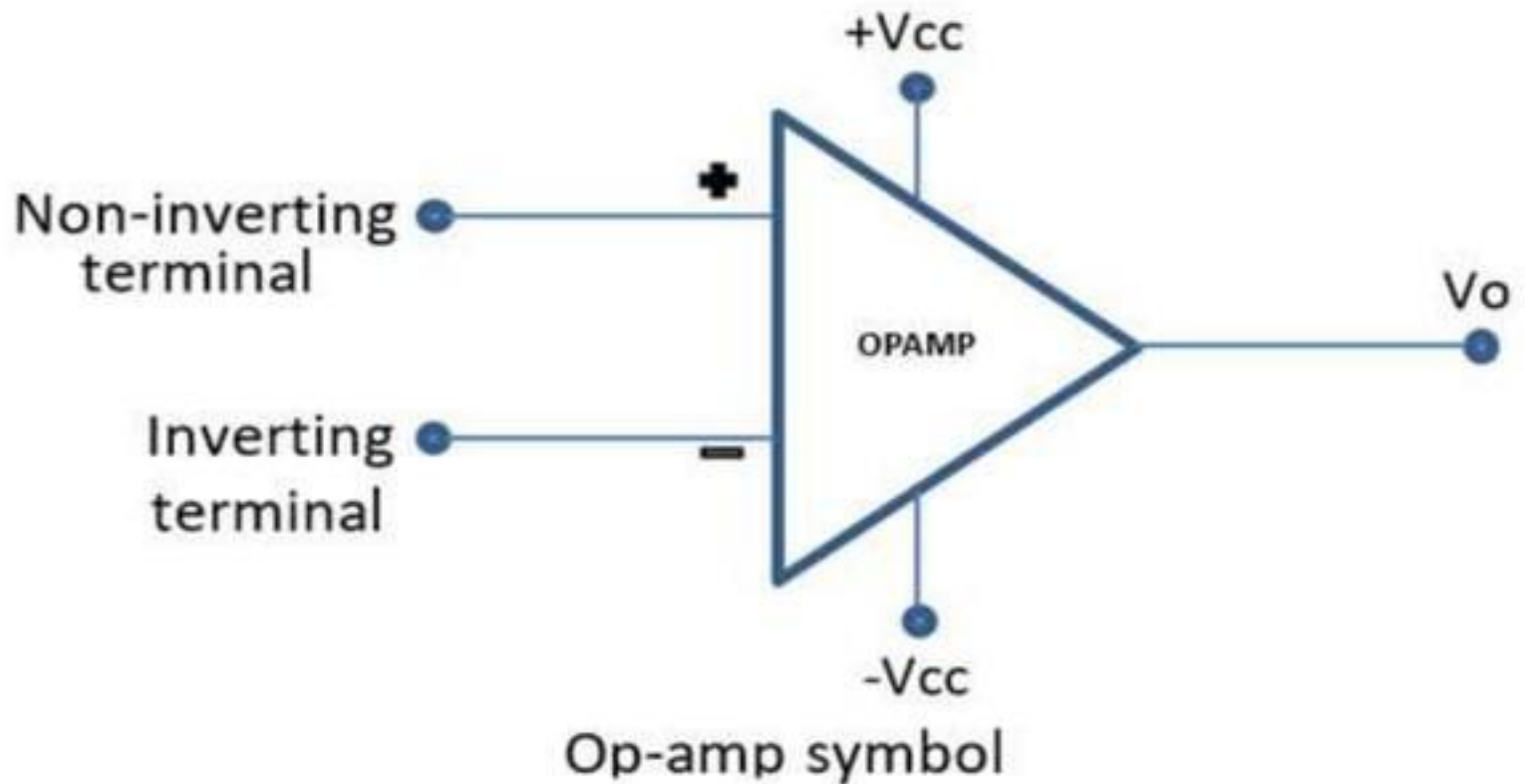
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- These feedback components determine the resulting function or “operation” of the amplifier and by virtue of the different feedback configurations whether resistive, capacitive or both, the amplifier can perform a variety of different operations, giving rise to its name of “Operational Amplifier”.

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- An *Operational Amplifier* is basically a three-terminal device which consists of two high impedance inputs. One of the inputs is called the **Inverting Input**, marked with a negative or “minus” sign, ($-$). The other input is called the **Non-inverting Input**, marked with a positive or “plus” sign ($+$).

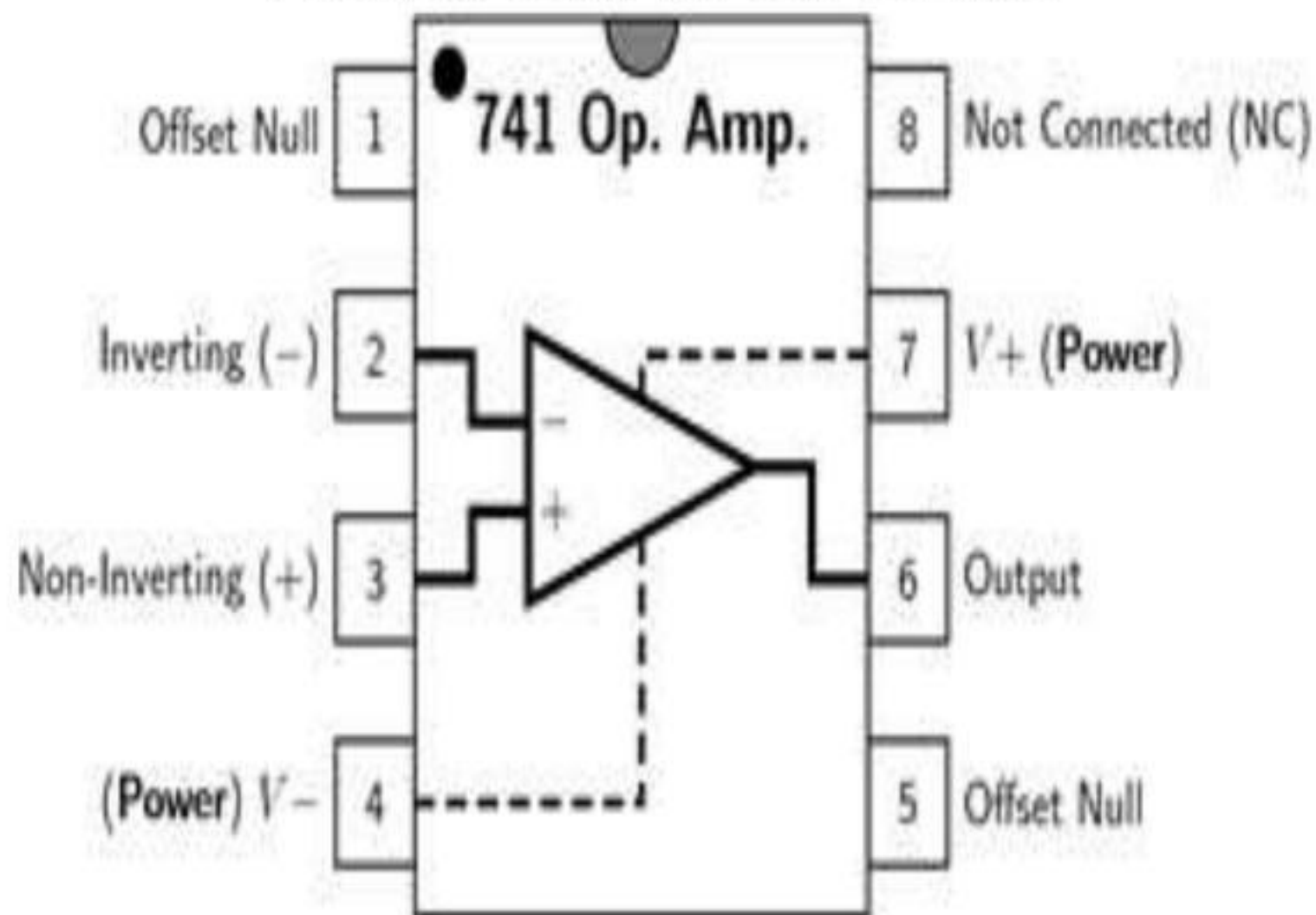
Symbol of OPAMP



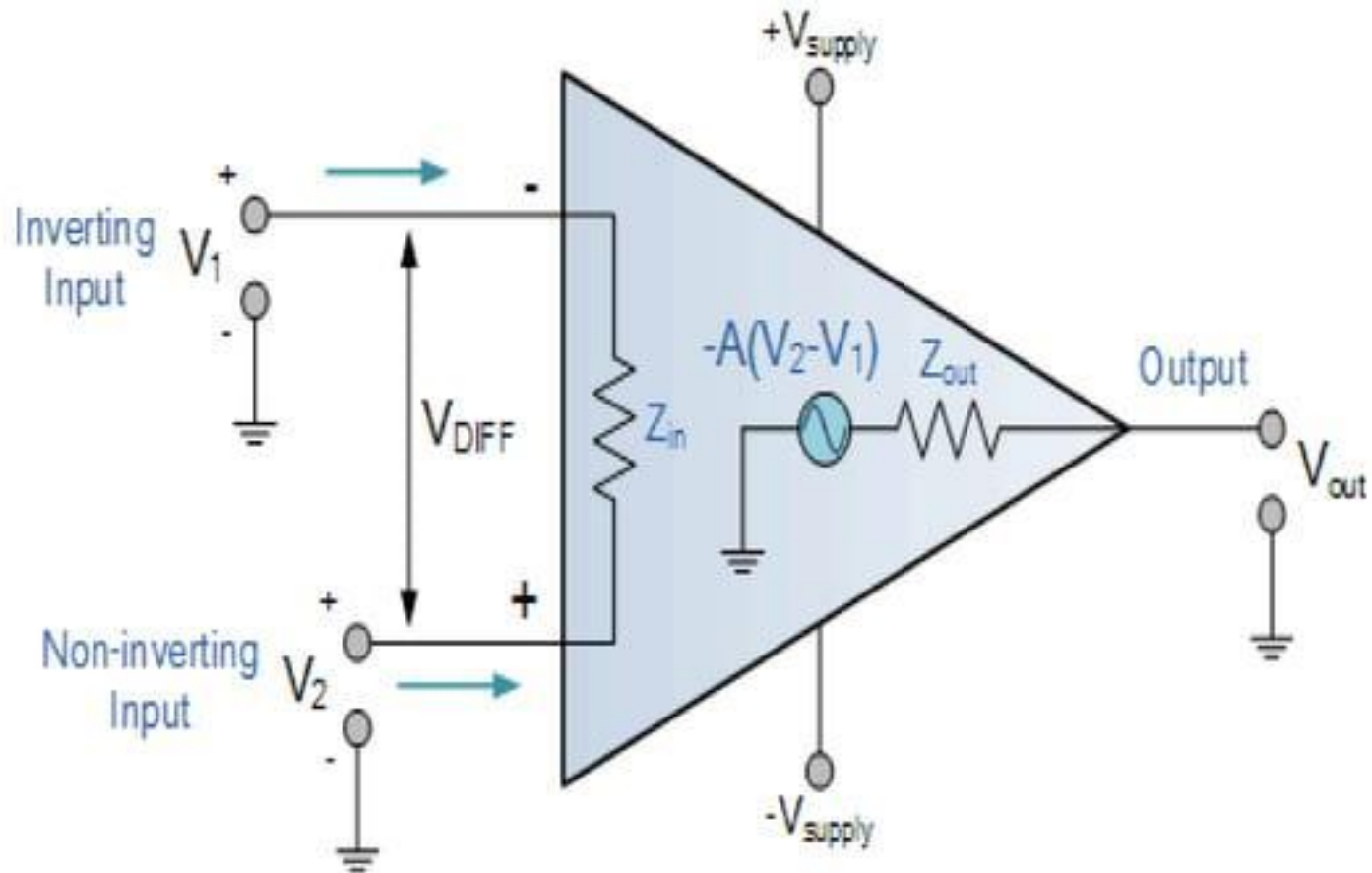
Four different classifications of operational amplifier gain.

- Voltage – Voltage “in” and Voltage “out”
- Current – Current “in” and Current “out”
- Transconductance – Voltage “in” and Current “out”
- Transresistance – Current “in” and Voltage “out”

PIN DIAGRAM OF OPMP

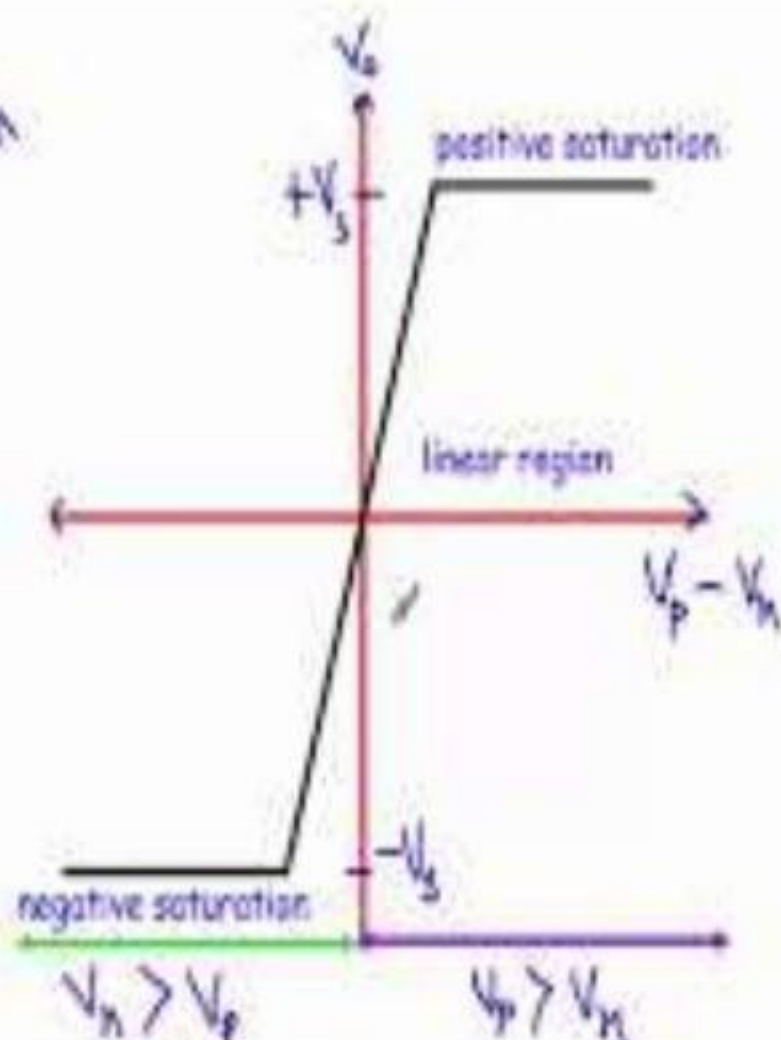
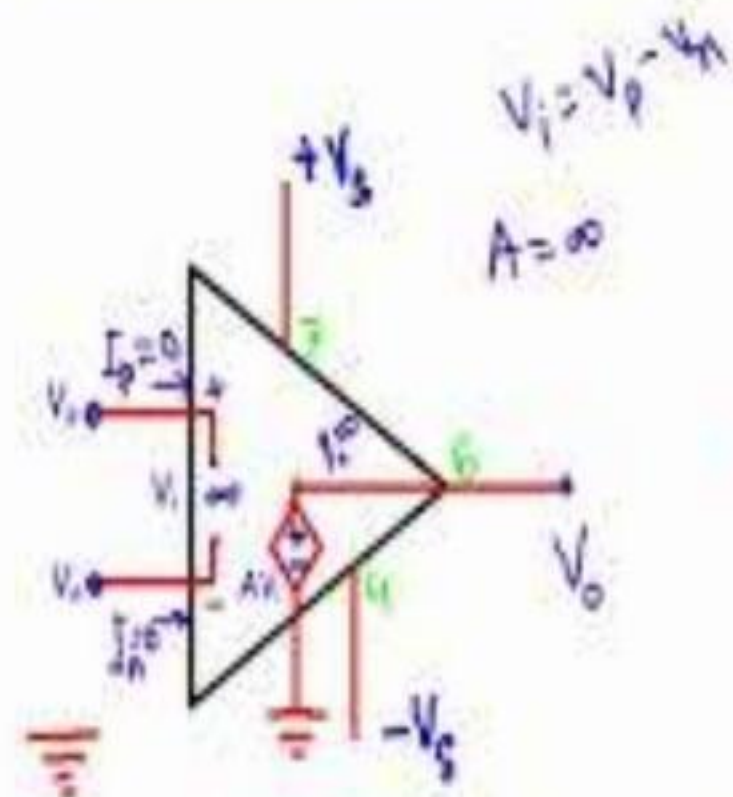


Equivalent Circuit of an Ideal Operational Amplifier



- $V_o = AV_{diff} = A (V_1 - V_2)$
- Which indicates that the output voltage V_o is a function of the difference between the input voltages V_1 and V_2 . For this reason op-amps are difference amplifiers.

Op-amp Transfer Characteristics



Non-Linear Region:

- An Op-amp operates in non-linear or saturation region when it is in open loop or positive feedback configuration.
- When $V_p > V_n$ by 1mA then opamp will be in positive saturation region $V_o = V_{sat} = +V_s$.
- When $V_n > V_p$ by 1mA then opamp will be in negative saturation region $V_o = -V_{sat} = -V_{supply}$

Linear Region:

- An Op-amp operates in linear when it is in negative feedback configuration.
- Current through inverting and non-inverting terminal is zero.
- inverting and non-inverting terminal voltages are equal.
- $V_o < |V_s|$ or $A(V_p - V_n) < |V_s|$
- $V_p - V_n < |V_s|/A$

V_s is 10V, 12V, 15V, $A = 1000000$ to 1000000000

Op-amp Parameter and Idealised Characteristic

- **Open Loop Gain, (A_{vo})**-the gain will be infinite but typical real values range from about 20,000 to 200,000.
- **Input impedance, ($Z_{IN-infinite}$)**-Input impedance is the ratio of input voltage to input current and is assumed to be infinite to prevent any current flowing from the source supply into the amplifiers input circuitry ($I_{IN} = 0$).

Real op-amps have input leakage currents from a few pico-amps to a few milli-amps.

Op-amp Parameter and Idealised Characteristic

- **Output impedance, ($Z_{OUT-Zero}$)**-The output impedance of the ideal operational amplifier is assumed to be zero acting as a perfect internal voltage source with no internal resistance so that it can supply as much current as necessary to the load.
- Real op-amps have output impedances in the 100-20k Ω range.

Op-amp Parameter and Idealised Characteristic

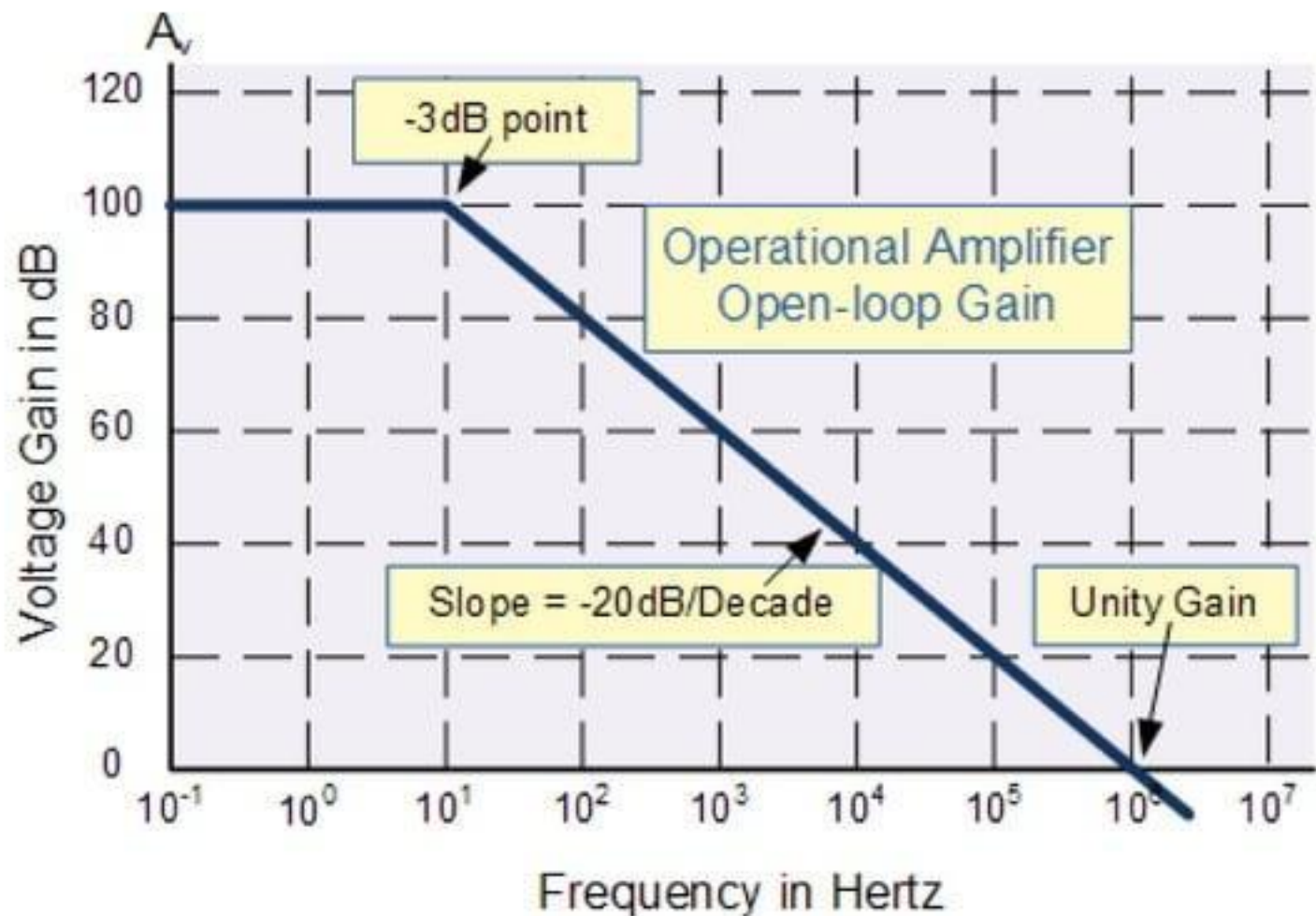
- **Infinite** – An ideal operational amplifier has an infinite frequency response and can amplify any frequency signal from DC to the highest AC frequencies so it is therefore assumed to have an infinite bandwidth.
- With real op-amps, the bandwidth is limited by the Gain-Bandwidth product (GB), which is equal to the frequency where the amplifiers gain becomes unity.

Op-amp Parameter and Idealised Characteristic

- **Offset Voltage, (V_{IO})**
- **Zero** – The amplifiers output will be zero when the voltage difference between the inverting and the non-inverting inputs is zero, the same or when both inputs are grounded.
- Real op-amps have some amount of output offset voltage.

- However, real **Operational Amplifiers** such as the commonly available **uA741**, for example do not have infinite gain or bandwidth but have a typical “Open Loop Gain” which is defined as the amplifiers output amplification without any external feedback signals connected to it and for a typical operational amplifier is about 100dB at DC (zero Hz).

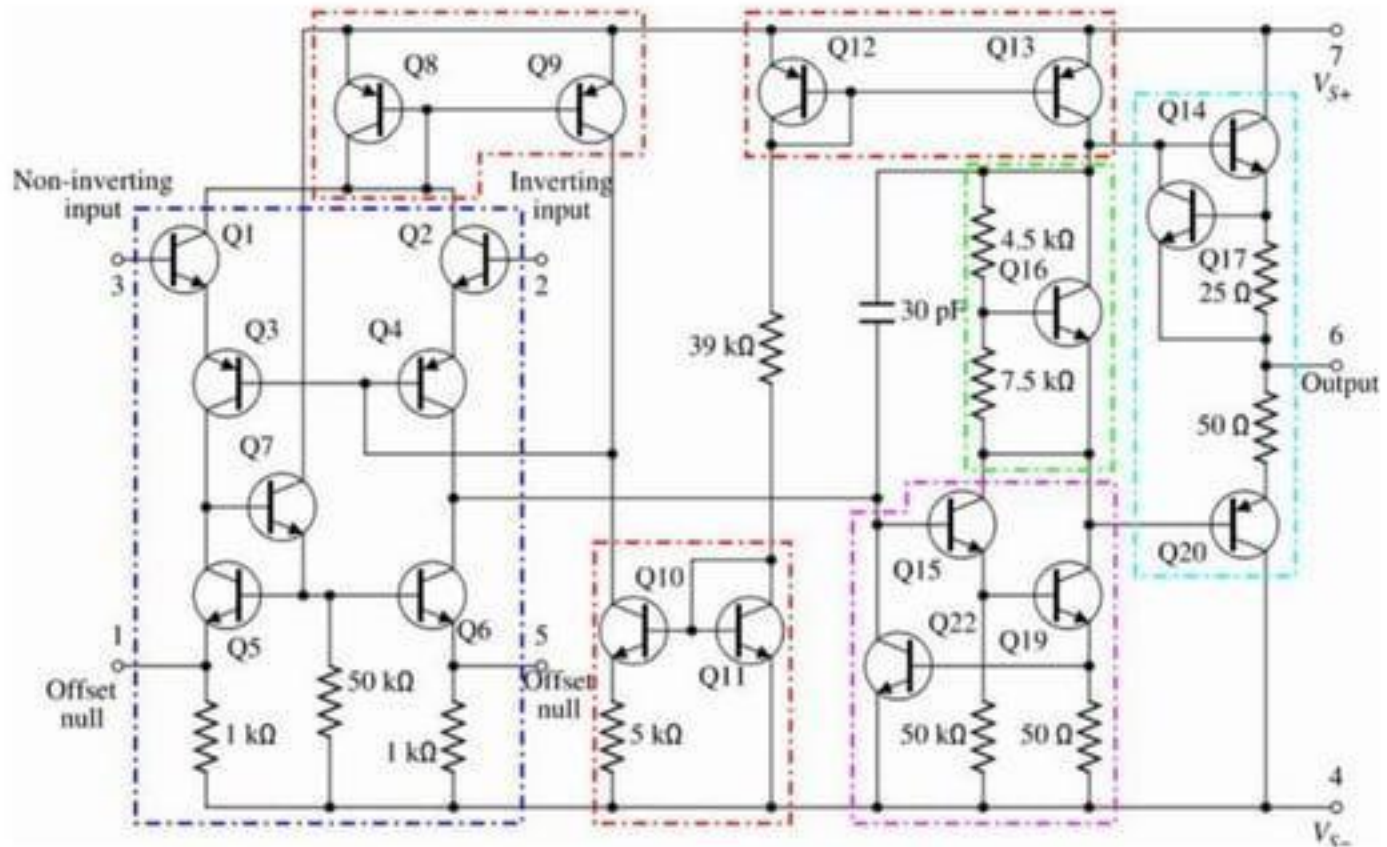
Open-loop Frequency Response Curve



Gain Bandwidth Product

- $GBP = \text{Gain} \times \text{Bandwidth} = A \times BW$
- Voltage Gain(A)= V_{out}/V_{in}
- In decibel $20\log(A)$
- An operational amplifiers bandwidth is inversely proportional to its gain, ($A \propto 1/BW$). Also, this -3dB corner frequency point is generally known as the “half power point”, as the output power of the amplifier is at half its maximum value

OPAMP Internal Diagram



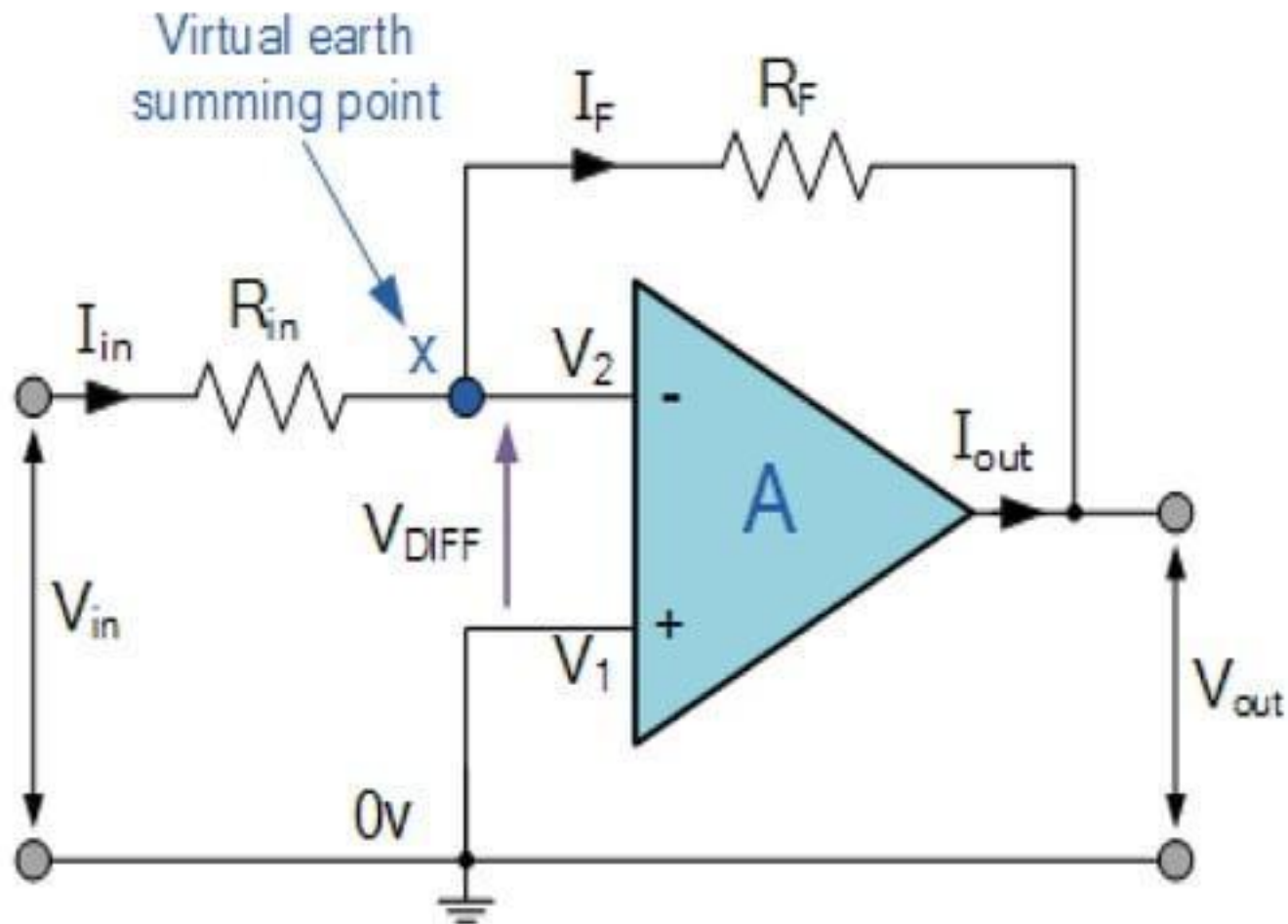
- A component-level diagram of the common 741 op-amp. Dotted lines outline:
- Current mirrors/current source (red)
- Differential amplifier (blue)
- Class A gain stage (magenta)
- Voltage level shifter (green);
- output stage (cyan)

- Like all op-amps, the circuit basically consists of three stages:
- **Differential amplifier** with high input impedance that generates a voltage signal, the amplified voltage difference .
- **Voltage amplifier** (class A amplification) with a high voltage gain to further amplify the voltage.
- **Output amplifier** (class AB push-pull emitter follower) with low output impedance and high current driving capability.

Summary

- We can connect external resistors or capacitors to the op-amp in a number of different ways to form basic “building Block” circuits such as, Inverting, Non-Inverting, Voltage Follower, Summing, Differential, Integrator and Differentiator type amplifiers.

Inverting OPAMP

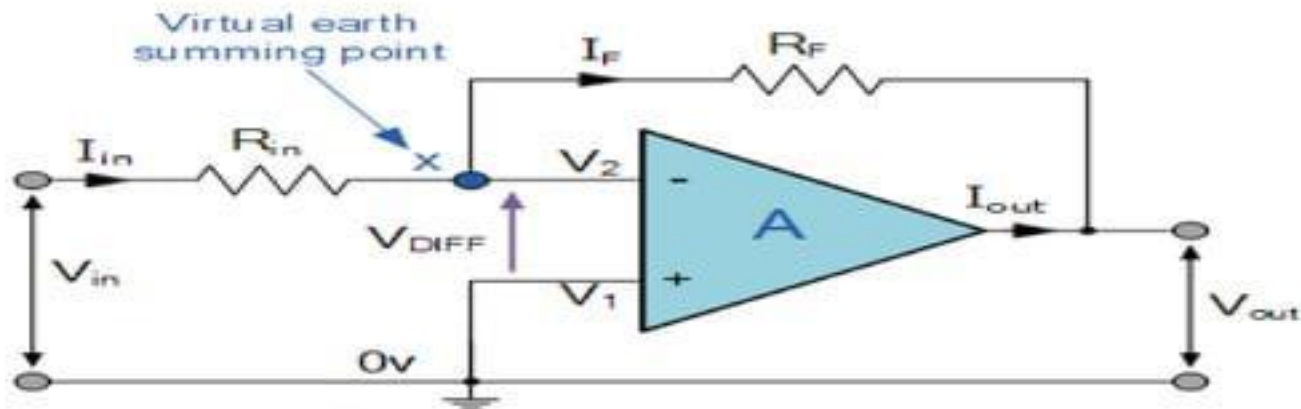


Inverting amplifier

- High gain is of no real use to us as it makes the amplifier both unstable and hard to control as the smallest of input signals.
- **Negative Feedback** is the process of “feeding back” a fraction of the output signal back to the input, but to make the feedback negative.
- We must feed it back to the negative or “inverting input” terminal of the op-amp using an external **Feedback Resistor** called R_f .
- This feedback connection between the output and the inverting input terminal forces the differential input voltage towards zero.

- This negative feedback results in the inverting input terminal having a different signal on it than the actual input voltage as it will be the sum of the input voltage plus the negative feedback voltage giving it the label or term of a **Summing Point**.
- We must therefore separate the real input signal from the inverting input by using an **Input Resistor**, R_{IN} .
- Feedback circuit results in the voltage potential at the inverting input being equal to that at the non-inverting input producing a *Virtual Earth* summing point because it will be at the same potential as the grounded reference input.
- In other words, the op-amp becomes a “differential amplifier”.

- Two very important rules to remember about **Inverting Amplifiers** or any operational amplifier for that matter and these are.
- No Current Flows into the Input Terminals
- The Differential Input Voltage is Zero as $V_1 = V_2 = 0$ (Virtual Earth)



$$i = \frac{V_{in} - V_{out}}{R_{in} + R_f}$$

$$\text{therefore, } i = \frac{V_{in} - V_2}{R_{in}} = \frac{V_2 - V_{out}}{R_f}$$

$$\text{so, } \frac{V_{in}}{R_{in}} = V_2 \left[\frac{1}{R_{in}} + \frac{1}{R_f} \right] - \frac{V_{out}}{R_f}$$

$$\text{and as, } i = \frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_f} \quad \frac{R_f}{R_{in}} = \frac{0 - V_{out}}{V_{in} - 0}$$

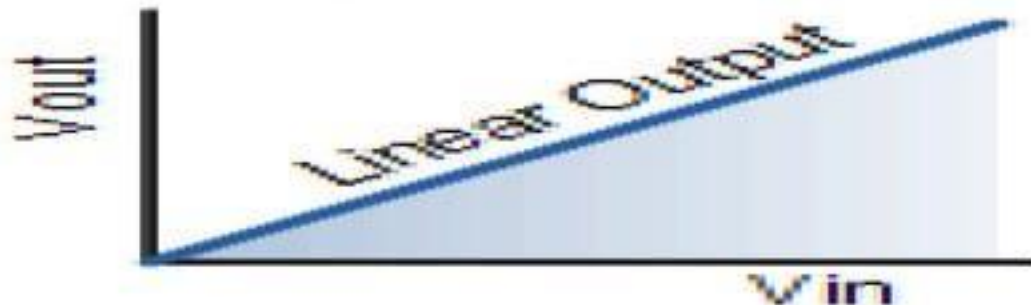
$$\therefore \frac{R_f}{R_{in}} = \frac{V_{out}}{V_{in}} \quad \therefore \frac{R_f}{R_{in}} = \frac{V_{out}}{V_{in}}$$

- Then, the **Closed-Loop Voltage Gain** of an Inverting Amplifier is given as.
- And this can be transposed to give V_{out} as:

$$\text{Gain } (A_v) = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

$$V_{out} = -\frac{R_f}{R_{in}} \times V_{in}$$

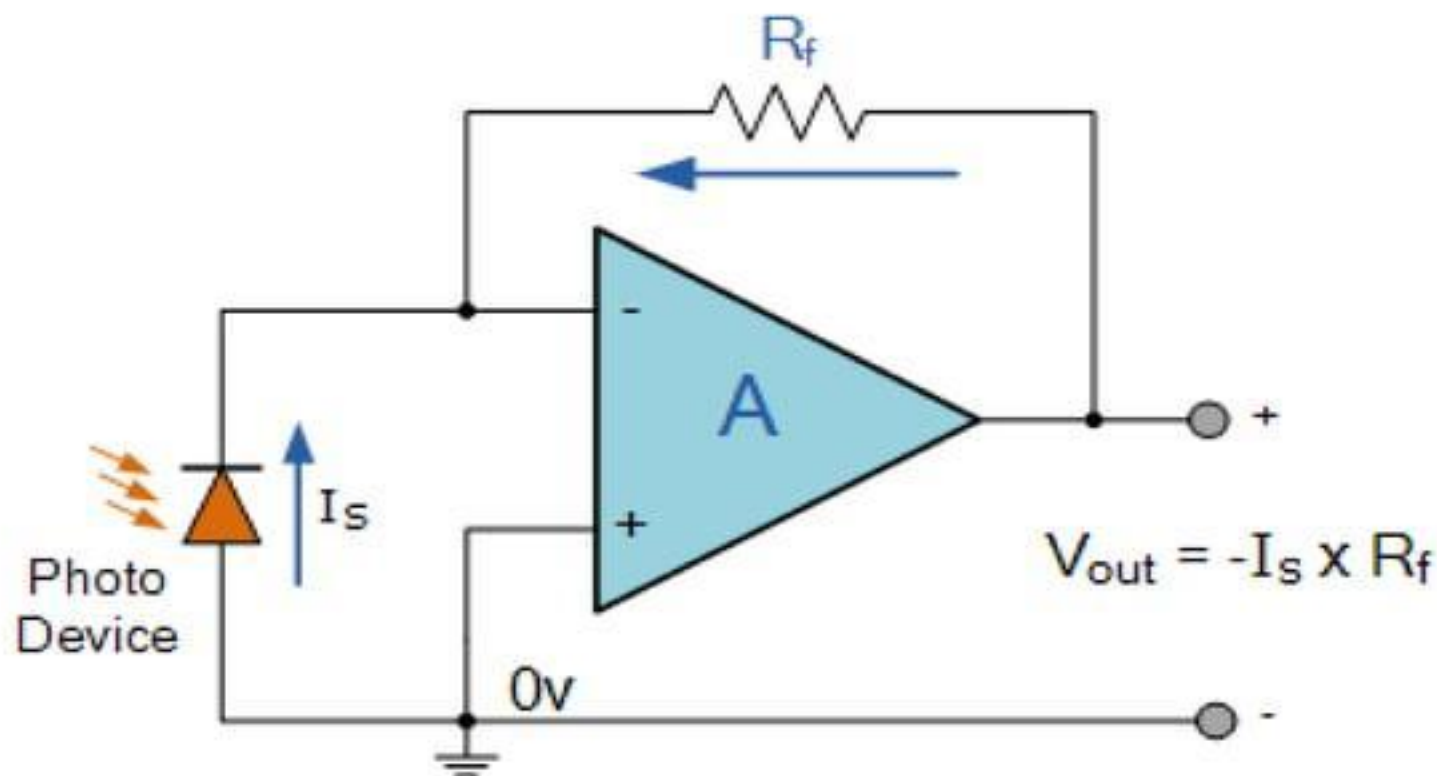
The negative sign in the equation indicates an inversion of the output signal with respect to the input as it is 180° out of phase. This is due to the feedback being negative in value.



$V_{out} = V_{in} \times \text{Gain}$. This property can be very useful for converting a smaller sensor signal to a much larger voltage.

- Another useful application of an inverting amplifier is that of a “transresistance amplifier” circuit.
- A **Transresistance Amplifier** also known as a “transimpedance amplifier”, is basically a current-to-voltage converter (Current “in” and Voltage “out”).
- They can be used in low-power applications to convert a very small current generated by a photo-diode or photo-detecting device etc, into a usable output voltage which is proportional to the input current as shown.

Transresistance Amplifier Circuit

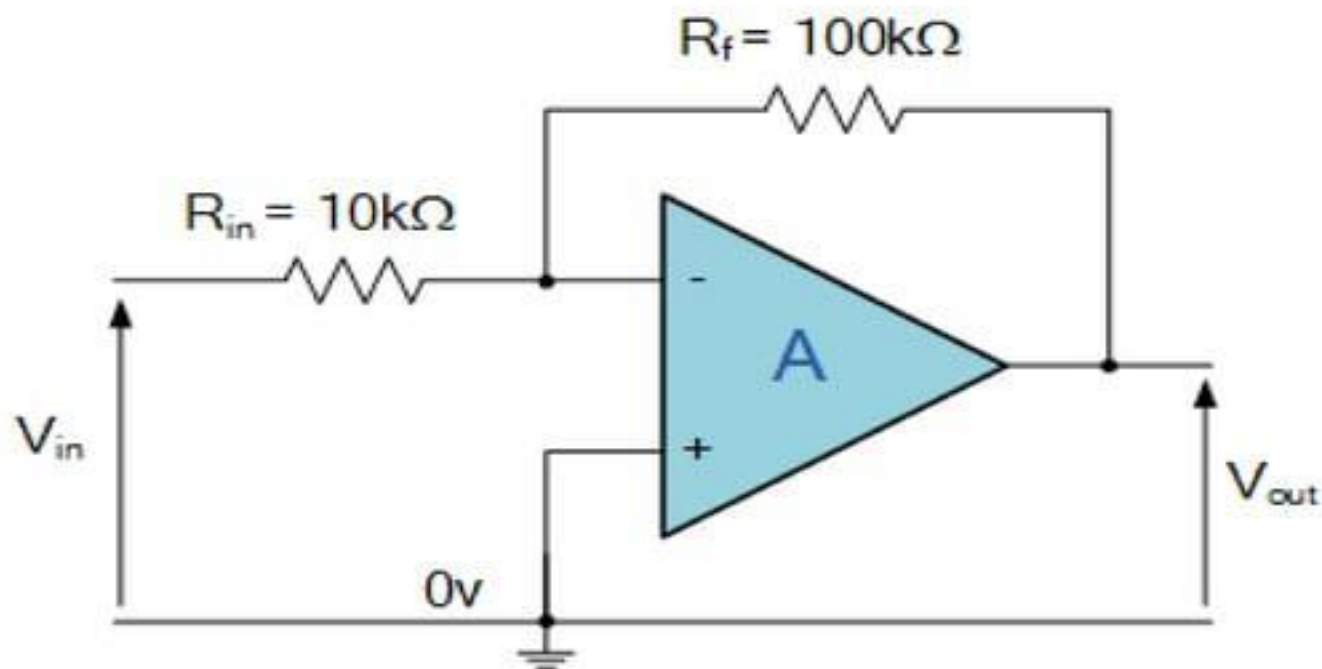


Transresistance Amplifier Circuit

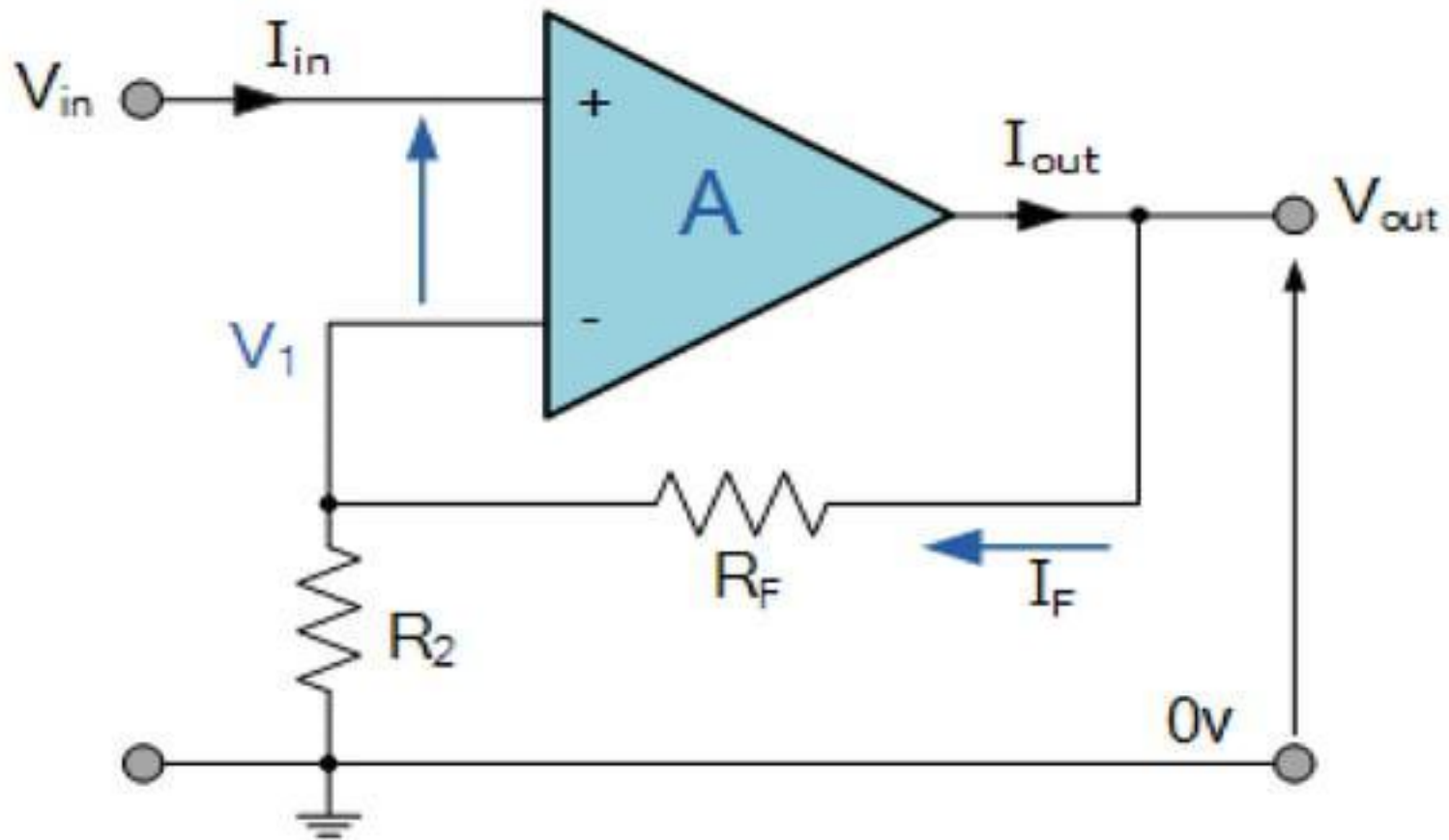
- The simple light-activated circuit above, converts a current generated by the photo-diode into a voltage.
- The feedback resistor R_f sets the operating voltage point at the inverting input and controls the amount of output.
- The output voltage is given as $V_{out} = -(I_s \times R_f)$. Therefore, the output voltage is proportional to the amount of input current generated by the photo-diode.

Example:1

- Find the closed loop gain of the following inverting amplifier circuit.

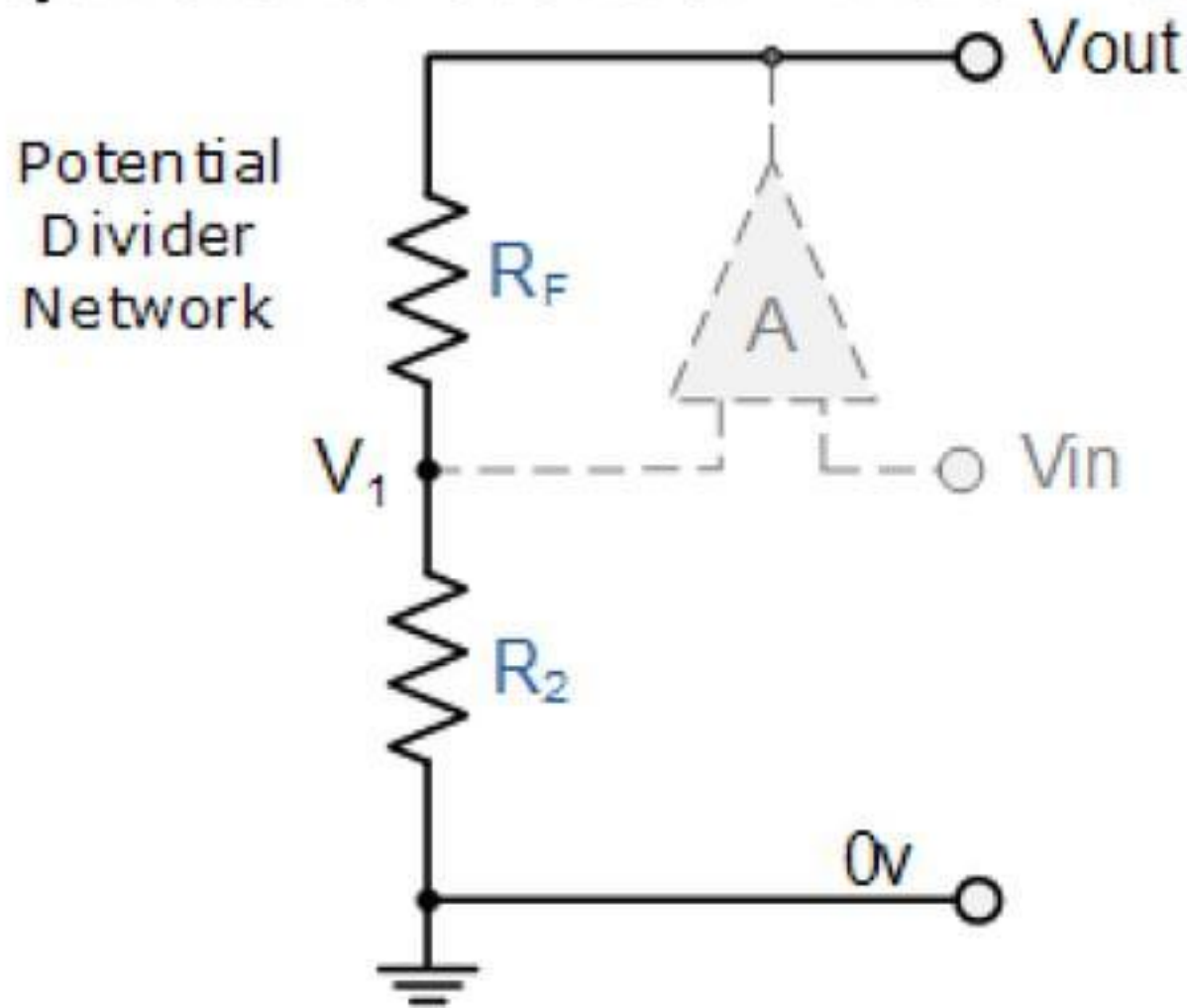


Non-Inverting Amp



- Feedback control of the non-inverting operational amplifier is achieved by applying a small part of the output voltage signal back to the inverting (–) input terminal via a R_f – R_2 voltage divider network, again producing negative feedback.
- High input impedance, R_{in} approaching infinity, as no current flows into the positive input terminal, (ideal conditions) and a low output impedance,

Equivalent Potential Divider Network



$$V_1 = \frac{R_2}{R_2 + R_F} \times V_{OUT}$$

Ideal Summing Point: $V_1 = V_{IN}$

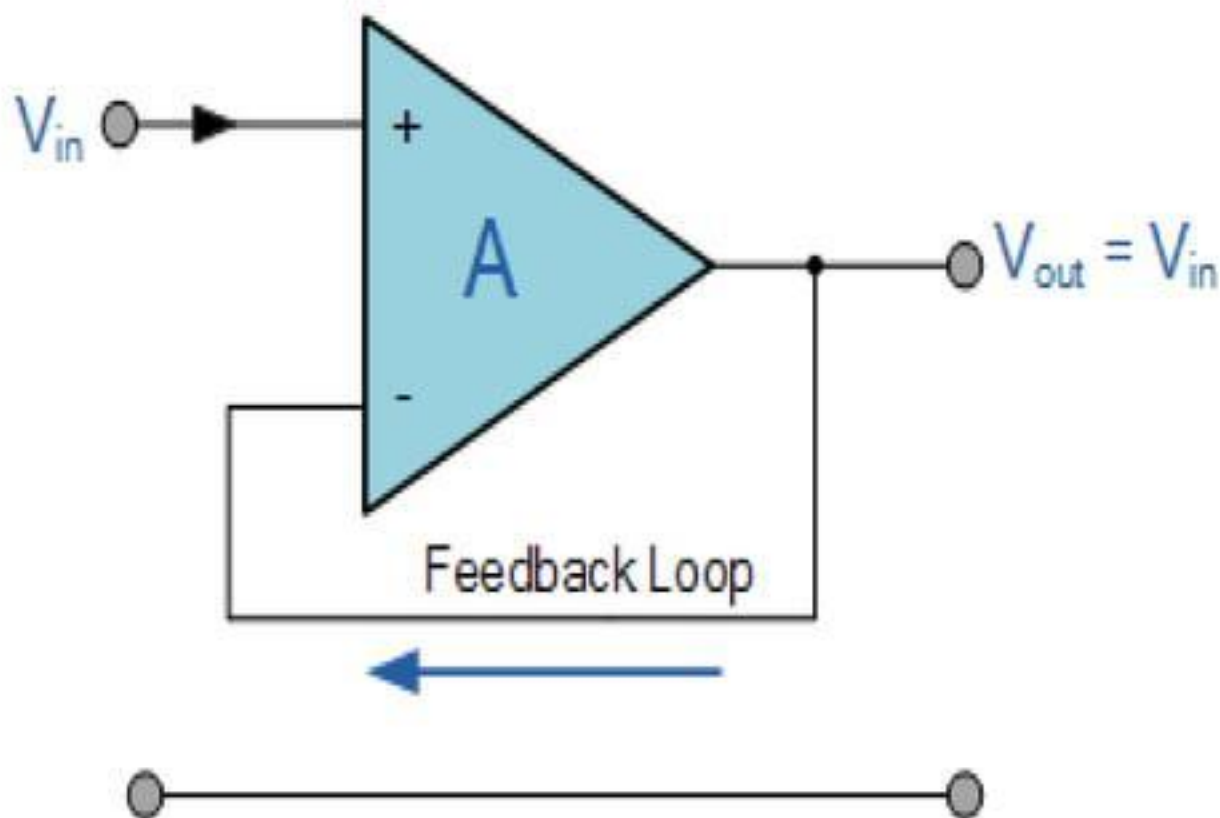
Voltage Gain, $A_{(V)}$ is equal to: $\frac{V_{OUT}}{V_{IN}}$

$$\text{Then, } A_{(V)} = \frac{V_{OUT}}{V_{IN}} = \frac{R_2 + R_F}{R_2}$$

$$\text{Transpose to give: } A_{(V)} = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_F}{R_2}$$

$$A_{(v)} = 1 + \frac{R_F}{R_2}$$

Non-Inverting as Voltage Follower



Voltage Follower

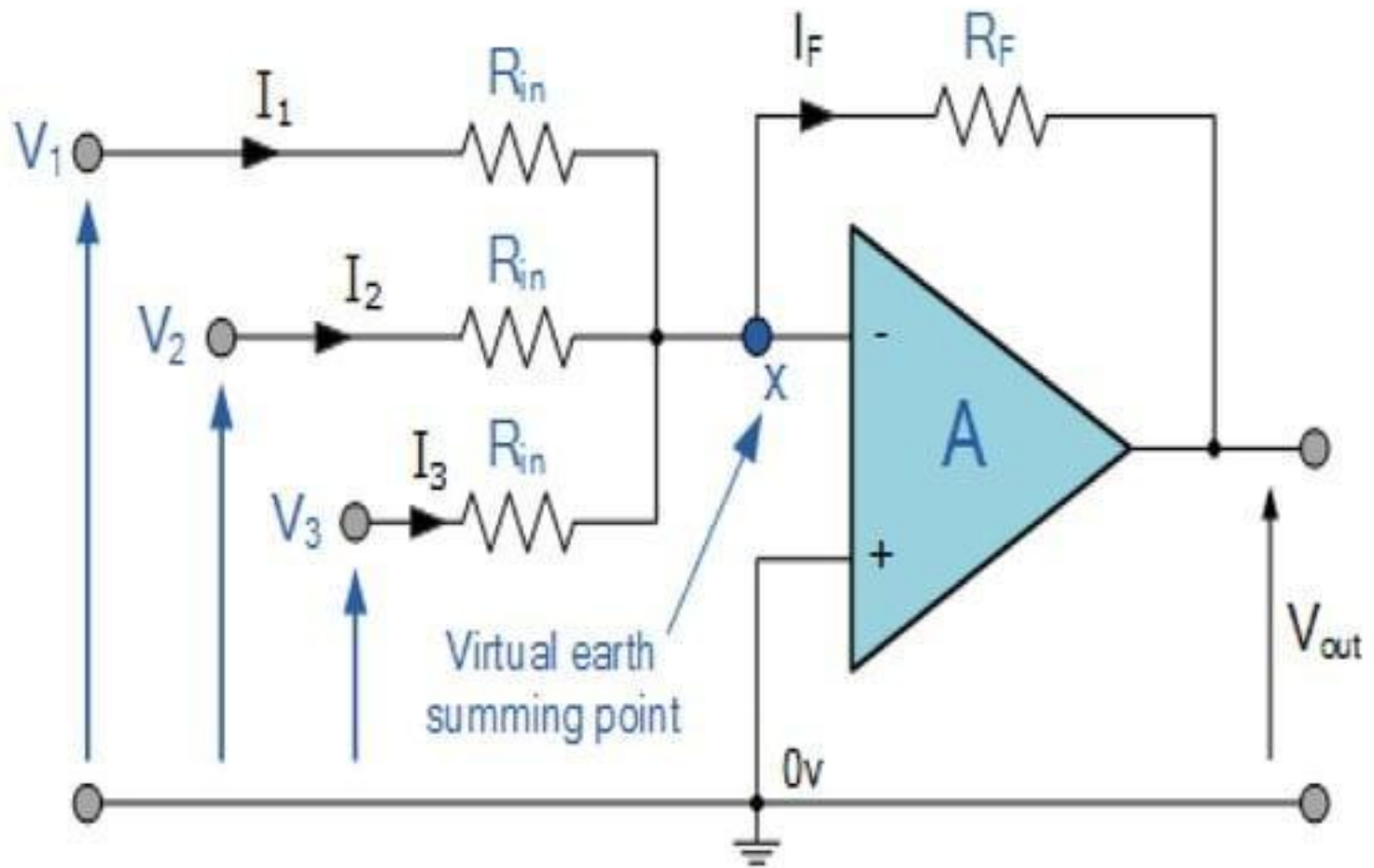
- In this non-inverting circuit configuration, the input impedance R_{in} has increased to infinity and the feedback impedance R_f reduced to zero.
- The output is connected directly back to the negative inverting input so the feedback is 100% and V_{in} is exactly equal to V_{out} giving it a fixed gain of 1 or unity.
- As the input voltage V_{in} is applied to the non-inverting input the gain of the amplifier is given as:

$$V_{\text{out}} = A(V_{\text{in}})$$

$$(V_{\text{in}} = V_{+}) \text{ and } (V_{\text{out}} = V_{-})$$

$$\text{therefore Gain, } (A_v) = \frac{V_{\text{out}}}{V_{\text{in}}} = +1$$

Summing Op-amp



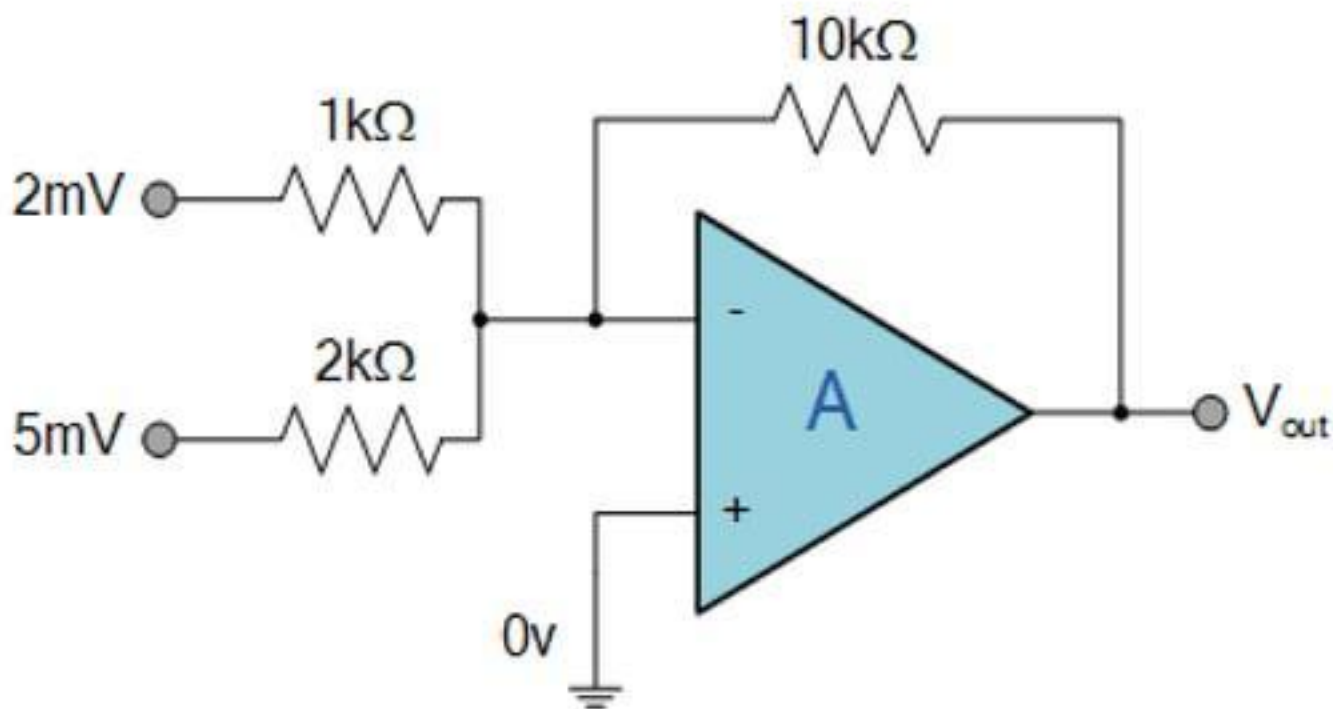
$$I_F = I_1 + I_2 + I_3 = - \left[\frac{V_1}{R_{in}} + \frac{V_2}{R_{in}} + \frac{V_3}{R_{in}} \right]$$

Inverting Equation: $V_{out} = -\frac{R_f}{R_{in}} \times V_{in}$

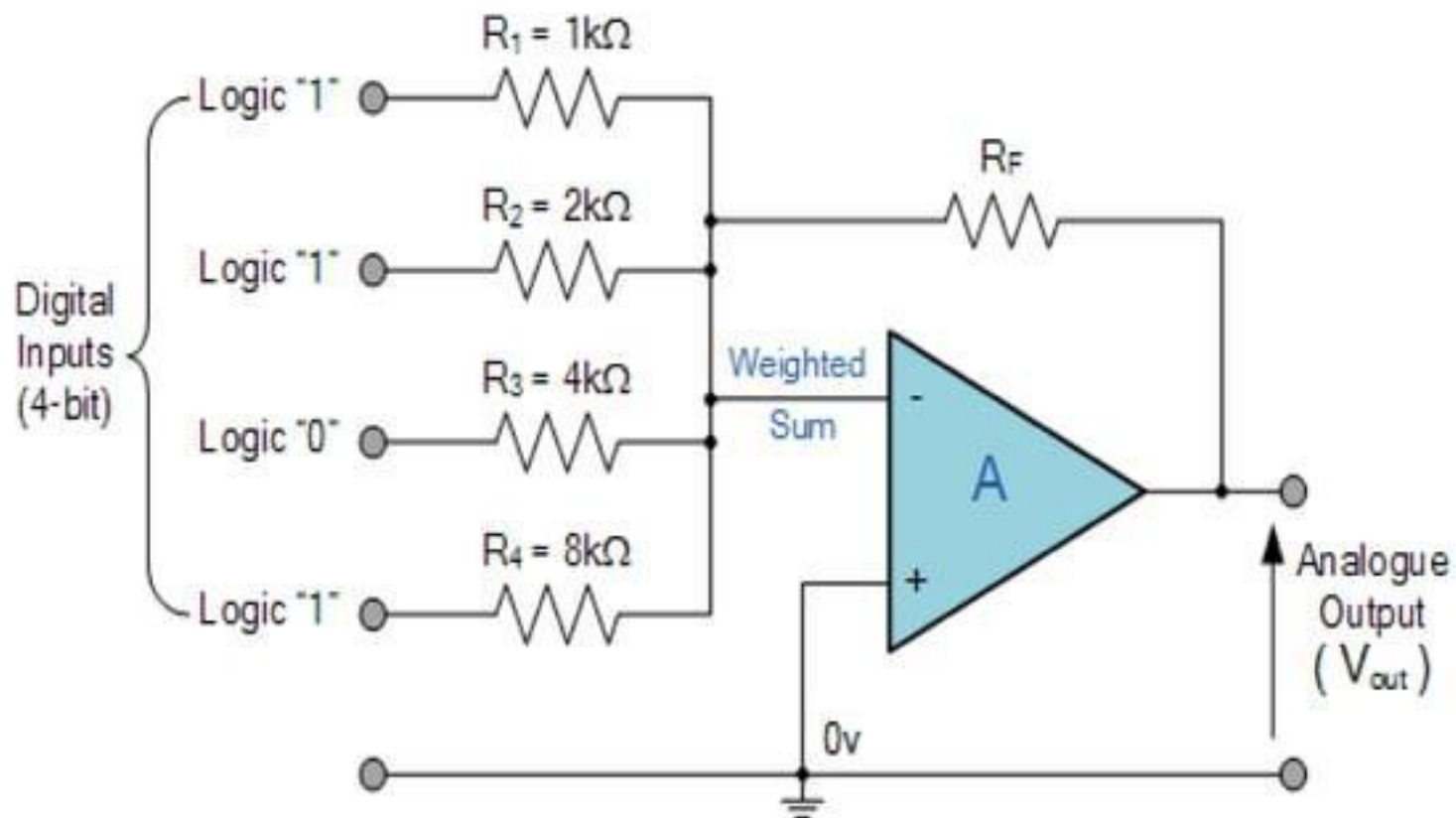
then, $-V_{out} = \left[\frac{R_F}{R_{in}} V_1 + \frac{R_F}{R_{in}} V_2 + \frac{R_F}{R_{in}} V_3 \right]$

Example No1

Find the output voltage of the following *Summing Amplifier* circuit.



Digital to Analogue Converter (DAC)



Summing Amplifier Audio Mixer

